# Lauryn E. Williams

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## Education

**University of Washington – Seattle** **Sep. 2022 – Jun. 2027**
*Doctor of Philosophy – Astronomy: Modeling the Formation and Evolution of Thorne-Żytkow Objects* *Seattle, WA*
*Master of Science – Astronomy* *Sep. 2022 – June 2024*

**University of Missouri – Columbia** **Aug. 2018 – May 2022**
*Bachelor of Science – Physics (emphasis in Astronomy)* *Columbia, MO*
*Bachelor of Science – Mathematics*

## Research Experience

**Department of Astronomy, University of Washington – Seattle:** **Sep. 2022 – Jun. 2027**
*Graduate Student*
*Seattle, WA*

- Studying the formation and evolution of Thorne-Żytkow Objects (TŻOs).
- Utilizing a variety of computational techniques, including binary population synthesis, moving-mesh hydrodynamics, and stellar evolution codes to understand the formation and complex interiors of TŻOs.

**Department of Physics and Astronomy, University of Missouri – Columbia:** **Feb. 2019 – May 2022**
*Undergraduate Research Assistant*
*Columbia, MO*

- Studied the morphology of young galaxies from the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey (CANDELS).
- Comparing and analyzing the physical parameter spaces of a variety of galaxies.

**American Museum of Natural History** **Jun. 2021 – Aug. 2021**
*Undergraduate Summer Researcher*
*(remote due to COVID-19)*

- Calculated the dynamical masses of companions from accelerating stars in the Hipparcos-Gaia Catalog of Accelerations (HGCA).
- Compared radial velocity curves of accelerating stars to determine the possibility of stellar companions.

**Boulder Solar Alliance, University of Colorado – Boulder:** **May 2020 – Aug. 2020**
*Undergraduate Summer Researcher*
*(remote due to COVID-19)*

- Focused on the Mid-Wave Infrared (MWIR) spectroscopic observations of the solar corona from the 2017 total solar eclipse.
- Produced detailed imaging of solar prominences and other active regions from comparing two different sets of coronal spectroscopic data.

## Publications

### Refereed

- **Lauryn E. Williams**, Philip Chang, Emily M. Levesque, Thomas R. Quinn *Neutron Star-Main Sequence Mergers Robustly Form Dynamically Stable Thorne-Żytkow Objects*, ApJ, 2025 ; DOI
- John Pharo, Yicheng Guo, Guillermo Barro Calvo, . . . , **Lauryn E. Williams**, Benjamin Winningham *The Dwarf Galaxy Population at $Z \sim 0.7$: A Catalog of Emission Lines and Redshifts from Deep Keck Observations*, ApJS, 2022 ; DOI
- Daniella C. Bardalez Gagliuffi, Jacqueline K. Faherty, Yiting Li, Timothy D. Brandt, **Lauryn E. Williams**, G. Mirek Brandt, Christopher Gelino *14 Her: A Likely Case of Planet-Planet Scattering*, ApJL, 2021 ; DOI

## Research Presentations

### Invited Talks

- Thursday Lunch Talks, *Princeton University*, **Oct. 2026**
- Transient Tea, ITC Frontiers, *Harvard–Smithsonian Center for Astrophysics*, **Oct. 2026**
- Coffee Talk Seminar, *Center for Computational Astrophysics*, **May 2026**

### Conference Presentations

*Modeling the Formation of Thorne-Żytkow Objects: Hydrodynamical Simulations of Neutron Star-Main Sequence Mergers*

- IAUS 402: Massive Stars Across Redshifts in the Era of JWST and Large-Scale Surveys | *Poster Presentation*, **Sep. 2025**
- AAS 246th Winter Meeting | *Oral Presentation*, **Jun. 2025**
- XXXII IAU General Assembly | *(virtual) Poster Presentation*, **Aug. 2024**

*Identifying Dynamical Masses Of Long Period Companions Using Orbit Fitting Code With HGCA Astrometry**

- Undergraduate Research and Creative Achievements Forum | *Poster Presentation*, **Apr. 2022**
- University of Missouri - Columbia Physics Colloquium | *(virtual) Oral Presentation*, **Oct. 2021**
- AMNH Physical Sciences Colloquium | *(virtual) Oral Presentation*, **Aug. 2021**

*Multi-Wavelength Identification of Extremely Young Galaxies with CANDELS**

- Mid-American Regional Astrophysics Conference (MARACS) | *(virtual) Oral Presentation*, **Apr. 2021**
- Undergraduate Research and Creative Achievements Forum | *(virtual) Oral Presentation*, **Apr. 2021**

*Comparisons of Novel Imaging and Spectroscopic Infrared Eclipse Observations in 2017**

- American Geophysical Union Conference (AGU) | *(virtual) Poster Presentation*, **Dec. 2020**
- University of Missouri - Columbia Astrophysics/Relativity Seminar | *(virtual) Oral Presentation*, **Oct. 2020**
- Solar Physics Division Conference | *(virtual) Poster Presentation*, **Aug. 2020**
- Boulder Solar REU Poster Walk | *(virtual) Poster Presentation*, **Aug. 2020**
- Boulder Solar REU Presentation Day | *(virtual) Oral Presentation*, **Jul. 2020**

**Undergraduate Projects*

## Teaching Experience

**Astronomy (ASTR101)** **Sep. 2024 – Dec. 2024**

*Graduate Teaching Assistant*

- Curriculum consisted of introductory conceptual and mathematical astronomy topics (modern theories, evolution of stars and galaxies, and astronomical distances, etc).
- Developed review material and lesson plans to lead 4 discussion sections (15 - 30 students each), held office hours throughout the week, and graded assignments and hands-on labs.

**The Planets (ASTR150)** **Jan. 2025 – Jun. 2025**

*Graduate Teaching Assistant*

- Curriculum included survey of the solar system, astronomical surveys and missions, and a small overview of exoplanet detection methods.
- Developed review material and lesson plans to lead 3 discussion sections (15 - 30 students each) for both the W25 and Sp25 academic quarters. Held office hours throughout the week, and graded assignments and hands-on labs. Prepared a full class lecture at the end of the quarter.

## Mentoring

### Current Students

Pre-Major in Astronomy Program (Pre-MAP) Student†, *Estimations of Dust Yields from AGB Stars within Dusty Star Forming Galaxies*

Kyleigh Medard, *Defining TŻO Formation Phases with Hydrodynamical Simulations*

### Former Students

Liam Becker, *Exploring the Effects of Neutron Star Velocities on TŻO Formation using Hydrodynamical Simulations*, (Feb. 2025 – Mar. 2026)

Pre-Major in Astronomy Program (Pre-MAP) Student†, *Identifying Thorne-Żytkow Object Candidates using APOGEE and Gaia*, (Oct. 2024 – Dec. 2024)

†*Anonymized for privacy*

## Outreach

**Public Talks**
Husky Giving Day, *Seattle, WA* | **Apr. 2026**
UW Alumni Association, *Seattle, WA* | **Apr. 2026**
Achievement Rewards for College Scientists (ARCS) Foundation, *Seattle, WA* | **Mar. 2026**
Sirius Astronomy Community Hour, *Seattle, WA* | **Apr. 2025**
You Grow Girl!, *Seattle, WA* | **Apr. 2025**
Future Fest, *Mill Creek, WA* | **Oct. 2022**

**UW Planetarium** **Sep. 2022 – Jun. 2026**
*Director* *Jun. 2025 – Jun. 2026*

- Introduced and installed new astrovisualization software (OpenSpace) and a robust media server into the planetarium. Led and trained 40+ volunteers to effectively create engaging planetarium shows. Delegated a variety of other management tasks to a team of 4 student coordinators to help give planetarium shows and special events to over 2000 public attendees.
- Organized and planned a variety of public multidisciplinary outreach events across the University of Washington's campus and the greater Seattle area.

*Volunteer Presenter* *Sep. 2022 – Jun. 2024*

- Received training to give interactive shows with the World Wide Telescope visualization software.
- Assisted other volunteers in presenting shows for special planetarium programming.

**WA NASA Space Grant Consortium** **Jun. 2023 – Aug. 2023**
*SURP Graduate Leader*

- Led and mentored a team of 5 undergraduates throughout the summer to prepare each for their individual poster presentation on their summer research.
- Provided professional development advice, research tips, and speaking opportunities within our weekly meetings.

**Maximizing Access to Research Careers (MARC)*** **Aug. 2020 – May 2022**
*Peer Mentor*

- Assisted 6 freshman, sophomore, and junior underrepresented minority students by meeting weekly to help discuss academic progress, answer questions, provide advice about campus life and research opportunities
- Attended weekly meetings with other peer mentors to plan MARC/IMSD meetings, activities, and speakers

**Undergraduate Experience*

## Honors & Awards

**Graduate Service/Outreach Award**, *University of Washington*, October 2026
**National Science Foundation (NSF) Graduate Research Fellowship**, April 2022
**Award for Academic Distinction**, *University of Missouri - Columbia*, April 2022
**Outstanding Graduating Senior**, *University of Missouri - Columbia*, April 2022

## Professional Memberships

- National Society of Black Physicists *(NSBP)*
- American Astronomical Society *(AAS)*